Fyodor Vinokurov

Backend / Full-Stack Engineer · Rust · Python · TypeScript

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Profiles

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Summary

Backend engineer (Rust, Python) who designs and ships complete systems solo — API, database, frontend and deployment. Largest project: a Rust platform with 237 endpoints and 1,000+ automated tests. 1st place at Decentrathon 5.0 as the team's sole developer; commercial QA experience at APARU with 100+ defect reports.

Experience

APARU LLP Jul 2026 – Aug 2026

QA Engineer & Technical Support

Manual QA and technical support for the APARU ride-hailing mobile app; data preparation for the ML team.

- Performed manual testing of the mobile app and filed 100+ defect reports covering UX/UI issues and backend failures (HTTP 500 errors, e.g. in the payment flow).
- Prepared datasets for the ML team and evaluated AI-powered feature outputs for incorrect or unreliable results.
- Provided technical support, reproducing and triaging user-reported issues.

Projects

Discord Bot Control Plane

Private · Multi-tenant platform that runs isolated Discord bots with a full ticket/support system, admin console, user dashboard and Discord Activity.

- Architected a Rust control plane that runs each bot as a systemd-isolated worker, reconciles desired runtime state and hands off secrets over a Unix socket authenticated with HKDF/HMAC, peer credentials and cgroup checks.
- Built a 237-endpoint REST/SSE API on 45 PostgreSQL migrations with Argon2 + TOTP + WebAuthn auth, AES-256-GCM secret storage and row/advisory locking for idempotent Discord side effects.
- Covered it with 794 Rust tests (356 against a real PostgreSQL) and 226 Vitest tests across 3 React apps; CI enforces fmt, lint, clippy (-D warnings), test and build.

RatScan — Static Malware Analysis for Minecraft Mods

Private · Rust API that statically analyses .jar mods for credential theft, download-and-execute chains and obfuscated loaders without executing them.

- Wrote a custom JVM class-file and bytecode parser with control-flow graphs and cross-method taint tracking, so a "malicious" verdict requires a proven chain (e.g. credential read → network send).
- Designed a detection engine with 81 YAML rules in 18 categories, dependency-scope classification and archive-bomb limits; verified by 76 unit tests and a 22-fixture regression corpus.
- Shipped async scan workers with SHA-256 deduplication, VirusTotal v3 enrichment cached in PostgreSQL and a supervised Telegram bot worker.

k0t1k — Agricultural Subsidy Scoring (Decentrathon 5.0 Winner)

Explainable merit-based scoring of agricultural subsidy applications for review committees.

- Built a 20-feature LightGBM model with per-applicant SHAP explanations and a weight-simulation tool, keeping final decisions with human reviewers; Russian/Kazakh outputs.
- Delivered a 38-endpoint FastAPI backend and 5-service Docker stack (PostgreSQL, Redis, Nginx, Cloudflare Tunnel) with JWT auth, rate limiting and 85 pytest tests.

https://github.com/hhhhhffd/dc-5.0-win-project

Pulse 109 — AI Triage for Public Services

GovTech Camp 2026 (BAITC / inDrive / Astana Hub) · Suggests topic, priority and recipient for citizen requests to the 109 service; analytics and forecasting for managers.

- Led architecture and ML for a 2-person team: defined Core/ML service boundaries and the PostgreSQL + Qdrant design for an 8-service stack.
- Built forecast backtesting on 83K real requests: Prophet reduced 30-day WAPE from 0.474 to 0.457, and a promotion gate kept the seasonal baseline at horizons where it performed better.

https://github.com/hhhhhffd/pulse109

LINKWATCH — Network Quality Monitoring

School internet-line monitoring: edge measurements feed incidents, provider tickets and verified restorations.

- Built a cross-platform (Windows/Linux) Rust edge agent with offline measurement buffering and retries.
- Developed the Go + PostgreSQL server with idempotent ingestion, incident tracking and provider-case workflows over 9 SQL migrations.

https://github.com/hhhhhffd/gov-ethernet-controller

DALES Student Portal

Private · Standalone replacement for a college LMS: scrapes schedule, grades, exams and forum into a fast web app with notifications.

- Built 6 typed HTML parsers behind a resilient upstream gateway (priority queue, single-flight, retries, circuit breaker, stale-cache fallback) and ~70 REST endpoints.
- Implemented an event store with outbox delivery to a Telegram bot (11 commands), HMAC-signed webhooks and iCalendar feeds; per-user credentials encrypted with AES-256-GCM.
- Tested with 102 automated tests on sanitized HAR-derived fixtures; deployed as a static musl binary under a hardened systemd unit.

Awards

1st Place — Decentrathon 5.0, GOV Case 2 — Apr 2026
BAITC
KZT 2M team prize. Sole developer of the winning subsidy-scoring system.

2nd Place — Code Masters Hackathon 4.0 — Oct 2025
Oskemen HUB
Team DataNomads.

3rd Place ×3 — Pizza Pitch & Idea Battle & DemoDay — Nov 2025
Oskemen HUB
Startup pitch competitions, team DataNomads.

Skills

Languages
Rust, Python, TypeScript, Go, SQL

Backend
Axum, Tokio, SQLx, FastAPI, SQLAlchemy, Alembic, Pydantic, WebSocket / SSE

Frontend
React 19, Vite, Tailwind CSS, TanStack Query

Data
PostgreSQL, Redis, SQLite, Qdrant

Security
Argon2, TOTP / WebAuthn, AES-GCM, HMAC, Rate limiting

ML & AI
LightGBM, SHAP, Prophet, NetworkX, OCR (RapidOCR / ONNX), LLM APIs

Testing & Quality
pytest, cargo test, Vitest, Playwright, clippy, mypy

Infrastructure
Nginx, systemd, Cloudflare Tunnel, GitHub Actions, Docker Compose, Linux

Education

Higher IT College, D. Serikbayev East Kazakhstan Technical University — Technical college diploma in Software Development
Sep 2024 – Jun 2028

Languages

Russian
Native

Kazakh
Intermediate (B1)

English
Pre-intermediate (A2–B1)